Endocrine Organ Lab Intro

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JeffMD-Endo/Uro/Repro Block

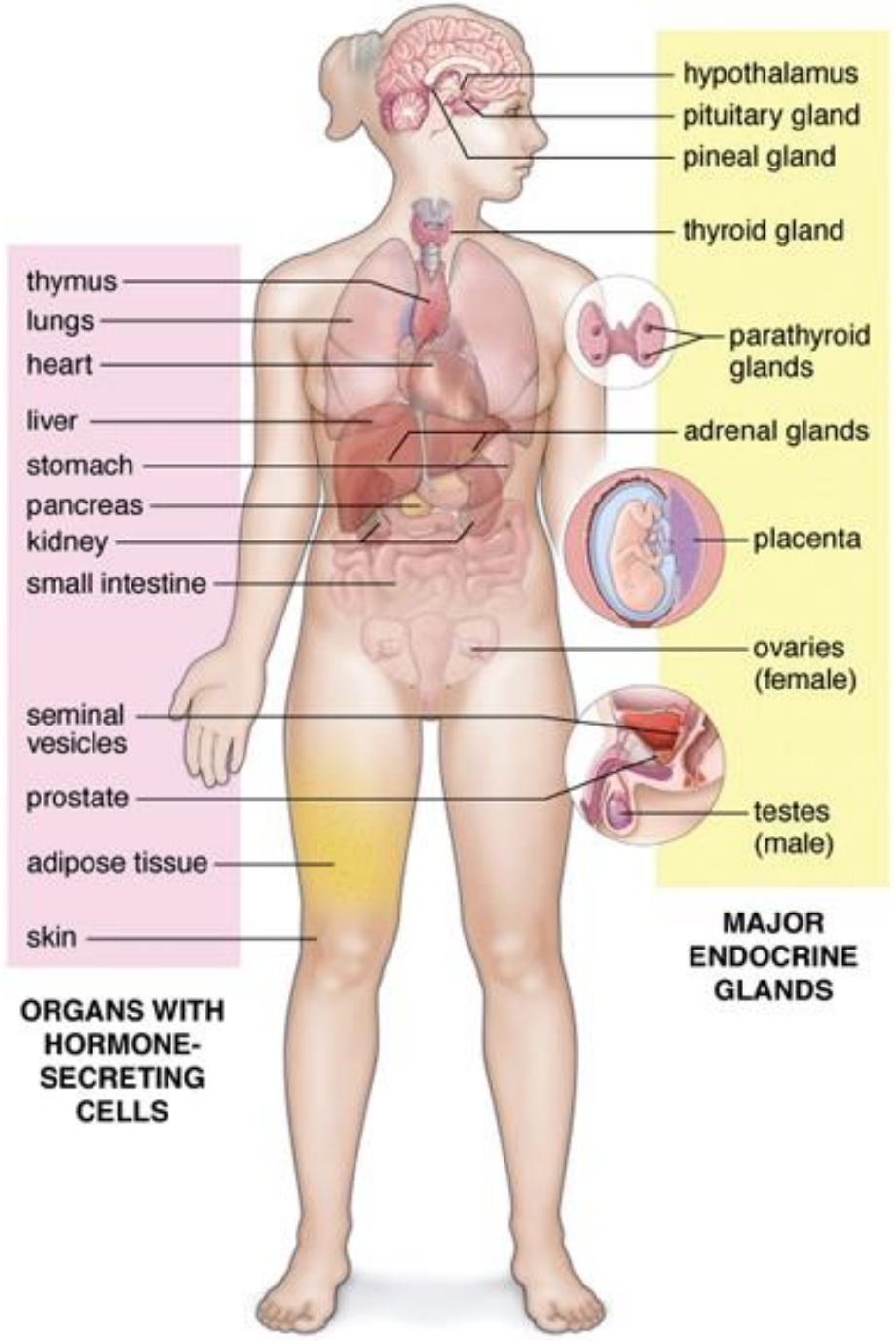
2026

Learning Objectives

1. Discuss the structural and functional anatomy of select organs in the endocrine system, including the hormones secreted by each and how they are moved into the blood supply
2. Identify the cells and tissues of select endocrine organs by histology
3. Compare and contrast the embryology of select endocrine organs and the impact of embryology on the adult function and structure at the gross and microscopic levels.
4. Predict how different pathologies of select endocrine organs might alter the histological or physiologic features of the component cells/tissues

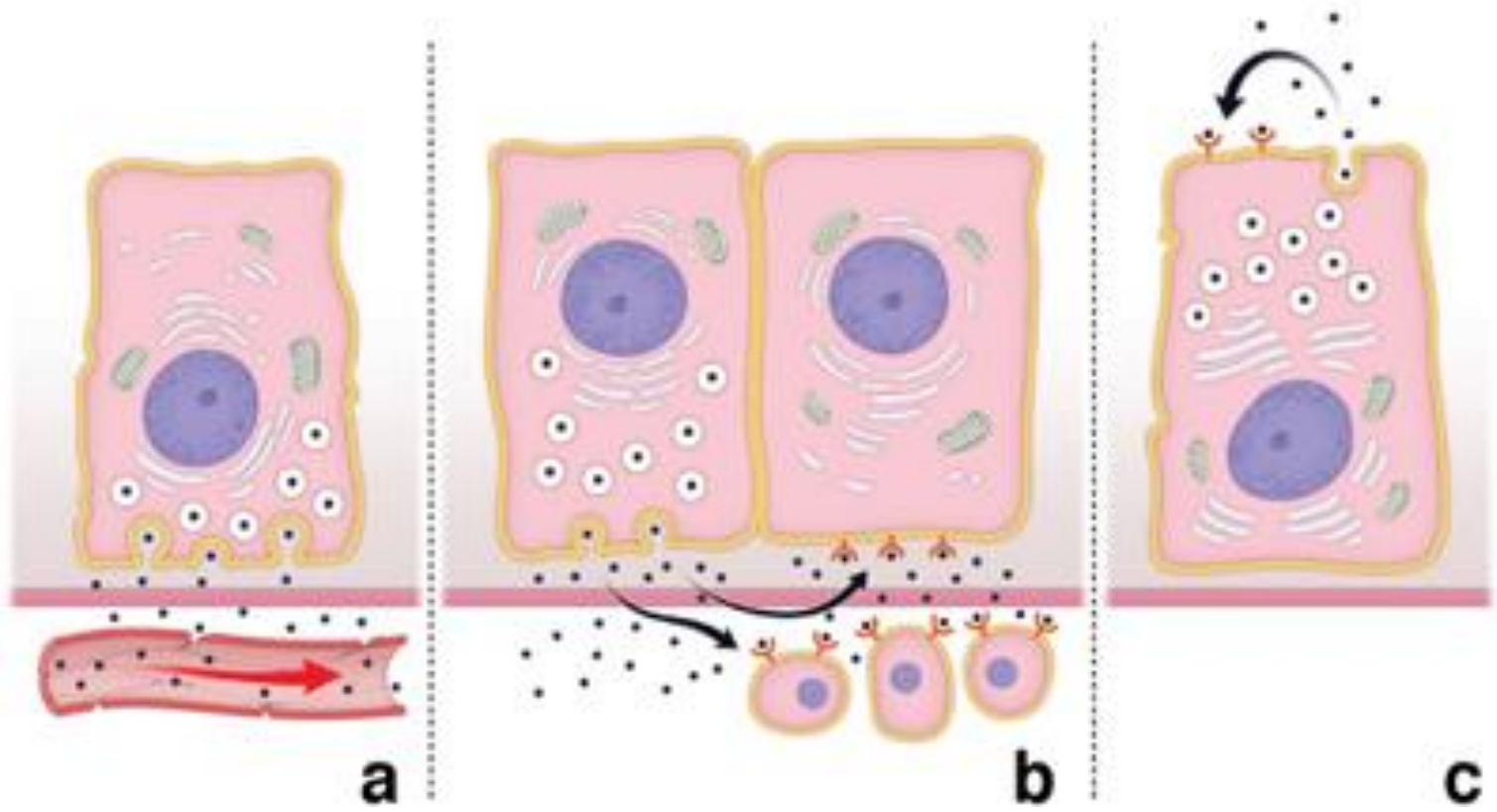
Endocrine System

- Complex system for controlling homeostasis
- Most organ systems have cells that secrete hormones for local control
  - DNES (Diffuse neuroendocrine system)
- Major organs exert more global effects



Methods of Communication

- Endocrine – global/systemic impact
  - Secretions into the blood to target multiple cell types
  - Via abundant fenestrated capillary networks
- Paracrine – local impact
  - Secretions into blood or interstitium to target adjacent tissues
- Autocrine – self regulation
  - Secretions feed back to the cells that made them to regulation function
- Not mutually exclusive!!



Signaling Molecules

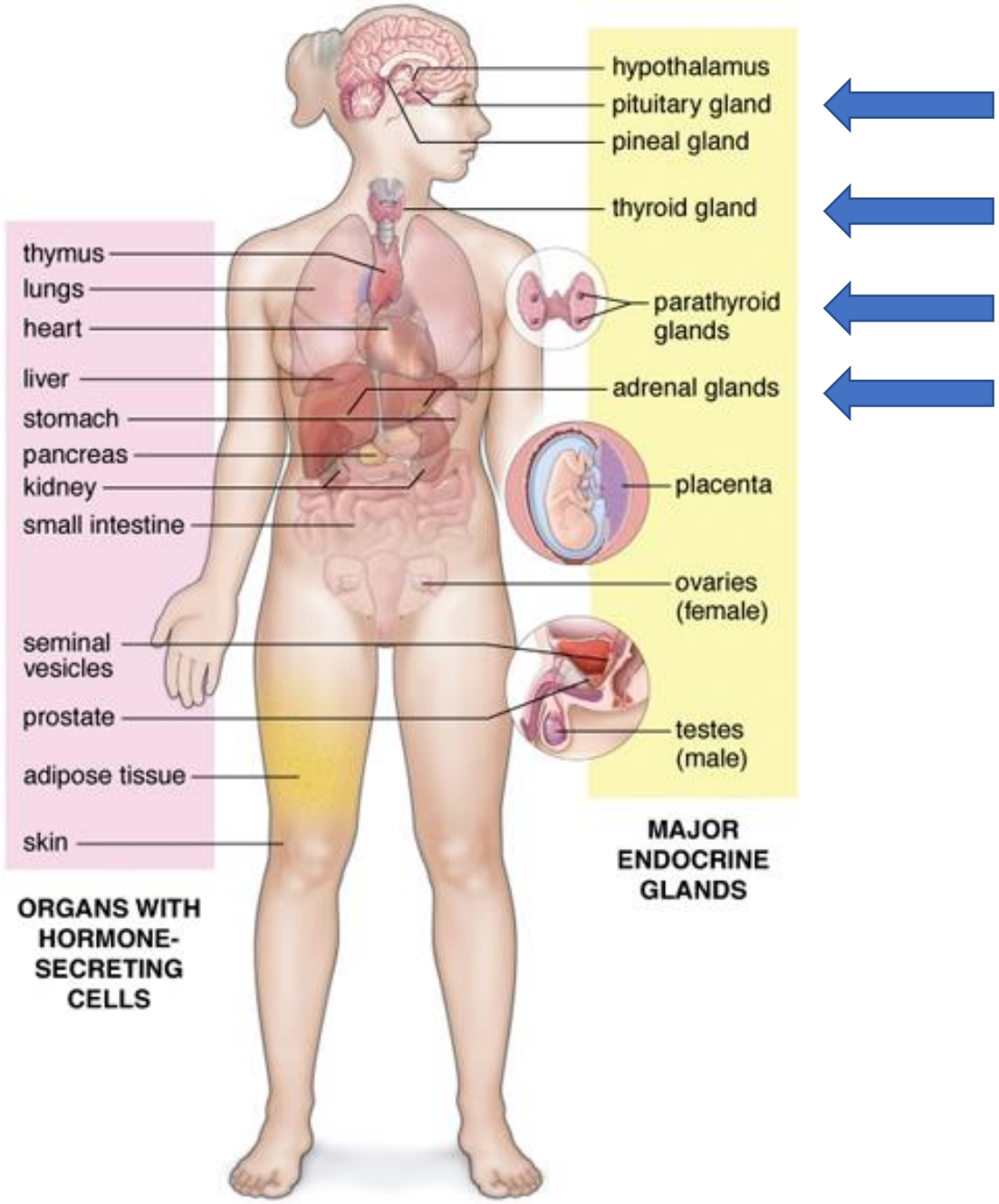
Secretions

- Peptides
- Proteins
- Steroids
- Neurotransmitters
- Lipid metabolites
- Amino acid derivatives

Target cells

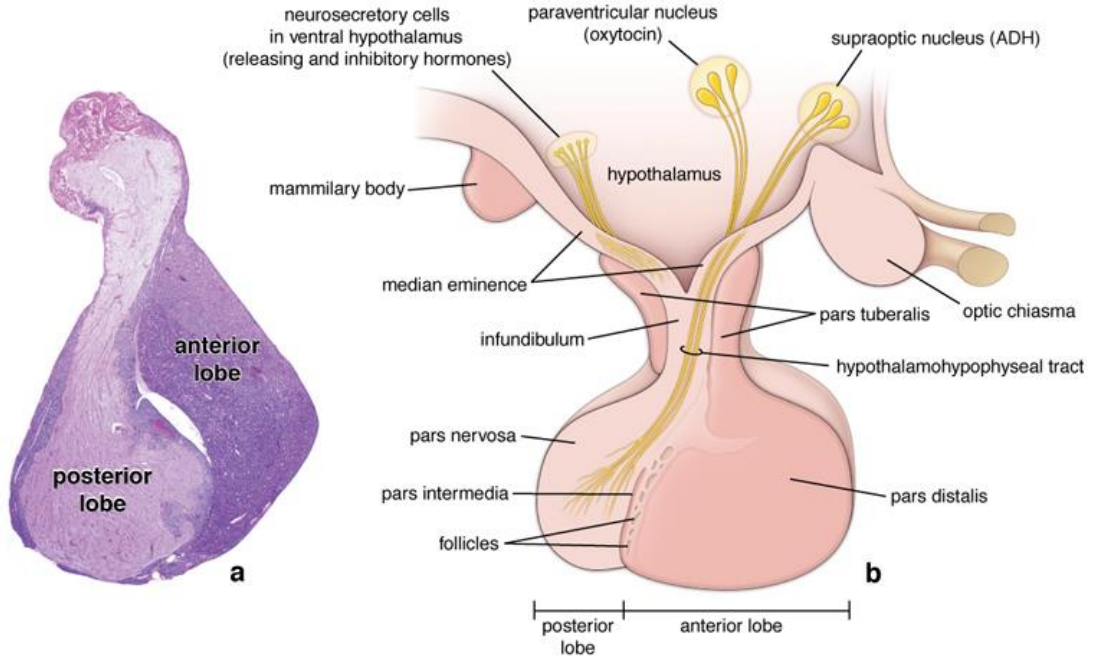
- Cell surface receptors
- Cytosolic receptors
- Intranuclear receptors

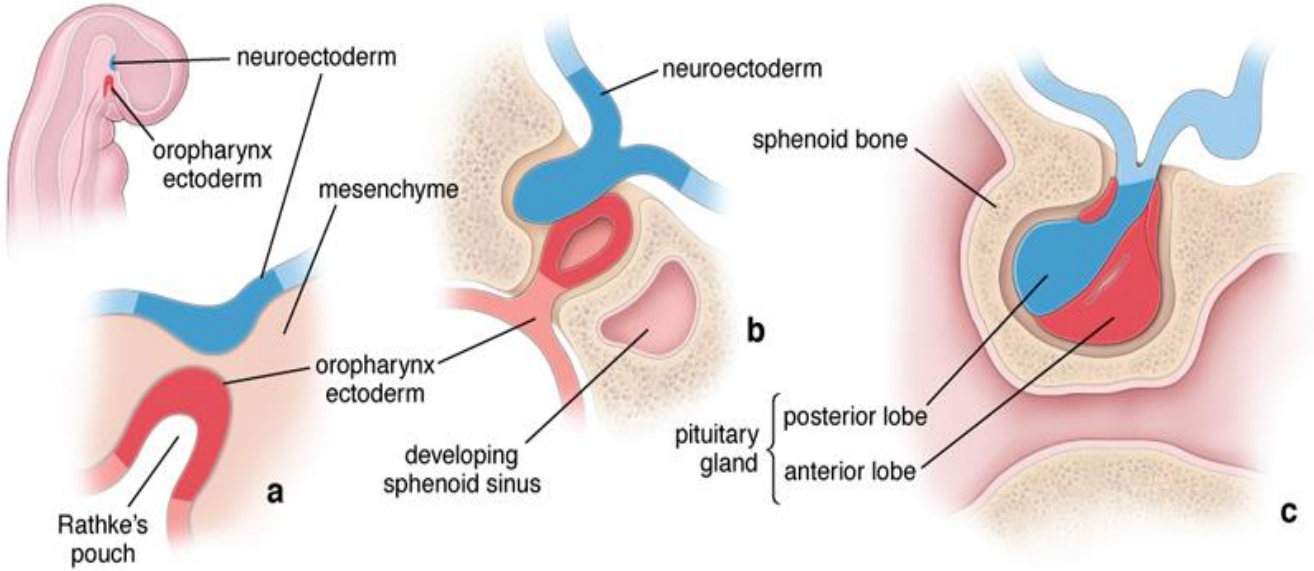
- Second messengers
- Kinase cascades
- Translational changes
- Transcriptional changes


hypothalamus
pituitary gland
pineal gland
thyroid gland
parathyroid glands
adrenal glands
placenta
ovaries (female)
testes (male)
MAJOR ENDOCRINE GLANDS
thymus
lungs
heart
liver
stomach
pancreas
kidney
small intestine
seminal vesicles
prostate
adipose tissue
skin
ORGANS WITH HORMONE-SECRETING CELLS

Pituitary Gland/ Hypophysis

- Master Gland
  - Two lobes from two different regions
- Anterior from ectoderm of oropharynx
  - Rathke's Pouch
- Posterior from neuroectoderm





Hormones Released From Each Lobe

Anterior

- Growth Hormone (Somatotropes)
- Prolactin (Lactotropes)
- Adrenocorticotropic hormone (Corticotropes)
- Follicle Stimulating hormone (Gonadotropes)
- Luteinizing Hormone (Gonadotropes)
- Thyroid stimulating hormone (Thyrotropes)

Posterior

- Oxytocin
- Antidiuretic hormone (ADH)

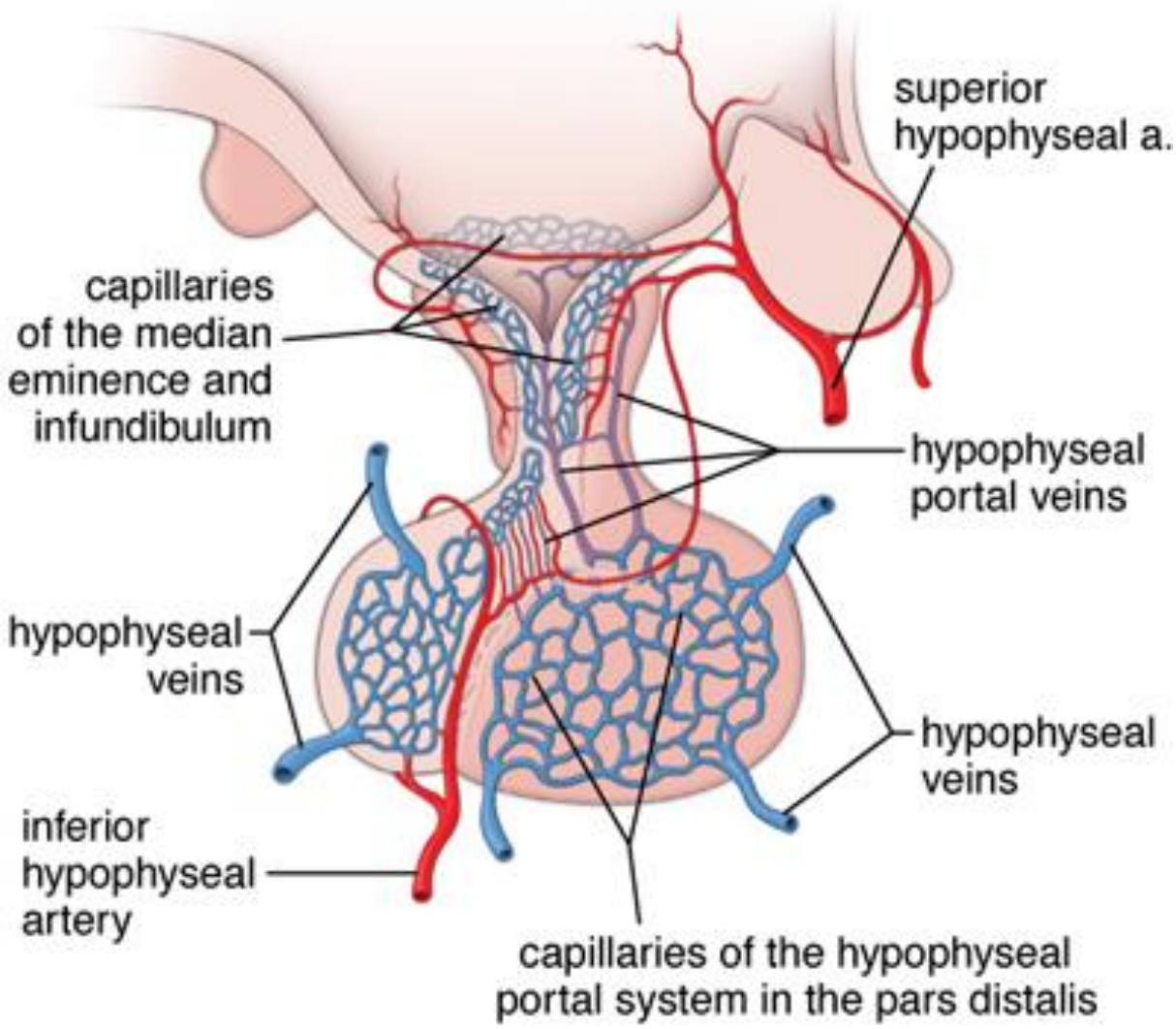
*** each are made by hypothalamic neurons and the axons extend into the Posterior Lobe and are released from there

Hypothalamic Control of Anterior Lobe Hormones

- Secretions from the anterior lobe are regulated by hormones/ Neurotransmitters made in the hypothalamus
- Growth hormone releasing hormone (GHRH)
- Somatostatin
- Corticotropin releasing hormone (CRH)
- Gonadotropin releasing hormone (GnRH)
- Thyrotropin releasing hormone (TRH)
- Dopamine (tuberoinfundibular pathway)

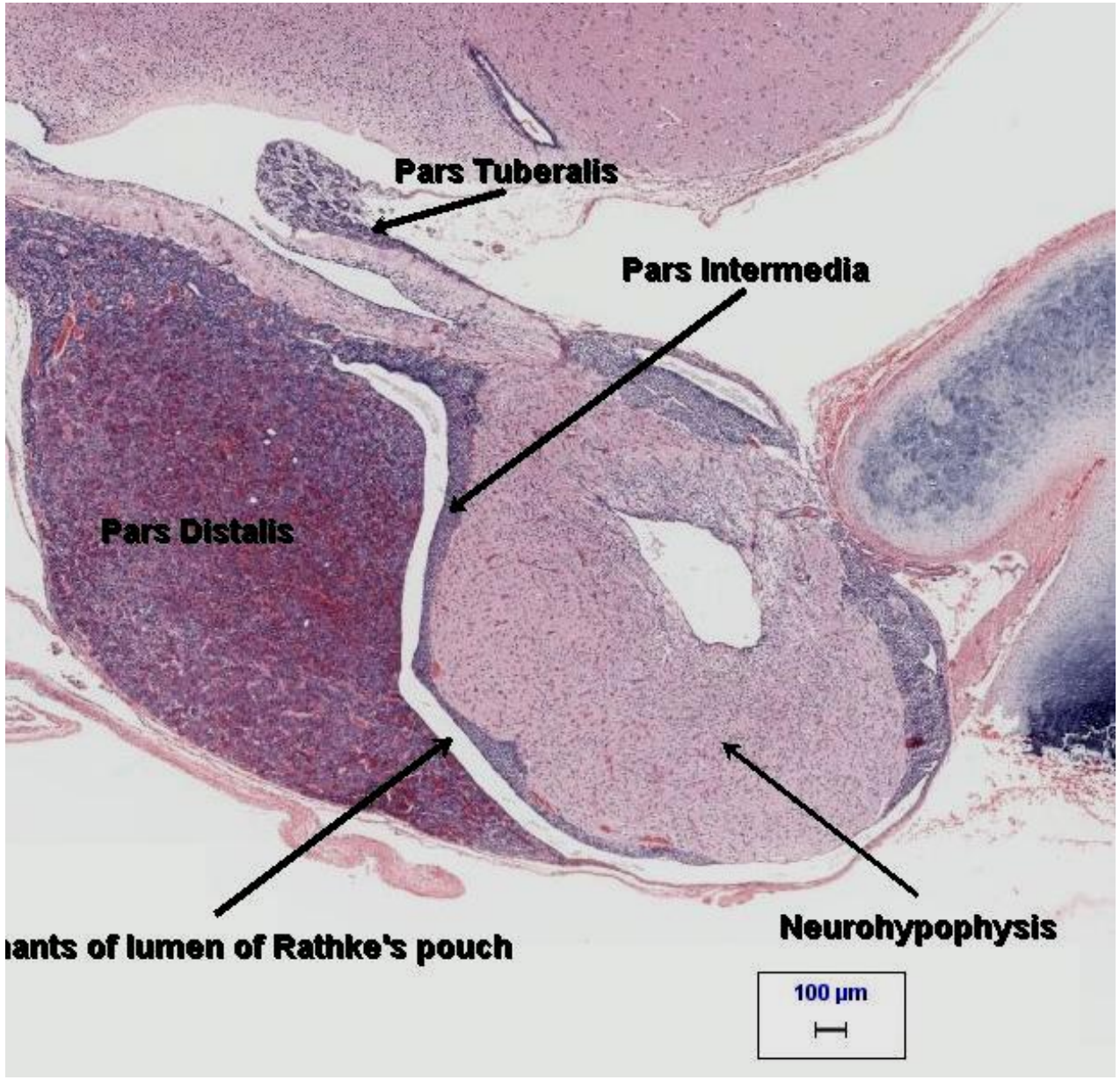
Hypophyseal Portal System

- Regulatory molecules are moved into the capillaries of the Hypothalamus
- Vessels form portal veins that move through infundibulum
  - Hypophyseal portal system
- Circulate through anterior lobe capillaries to control pituitary secretions



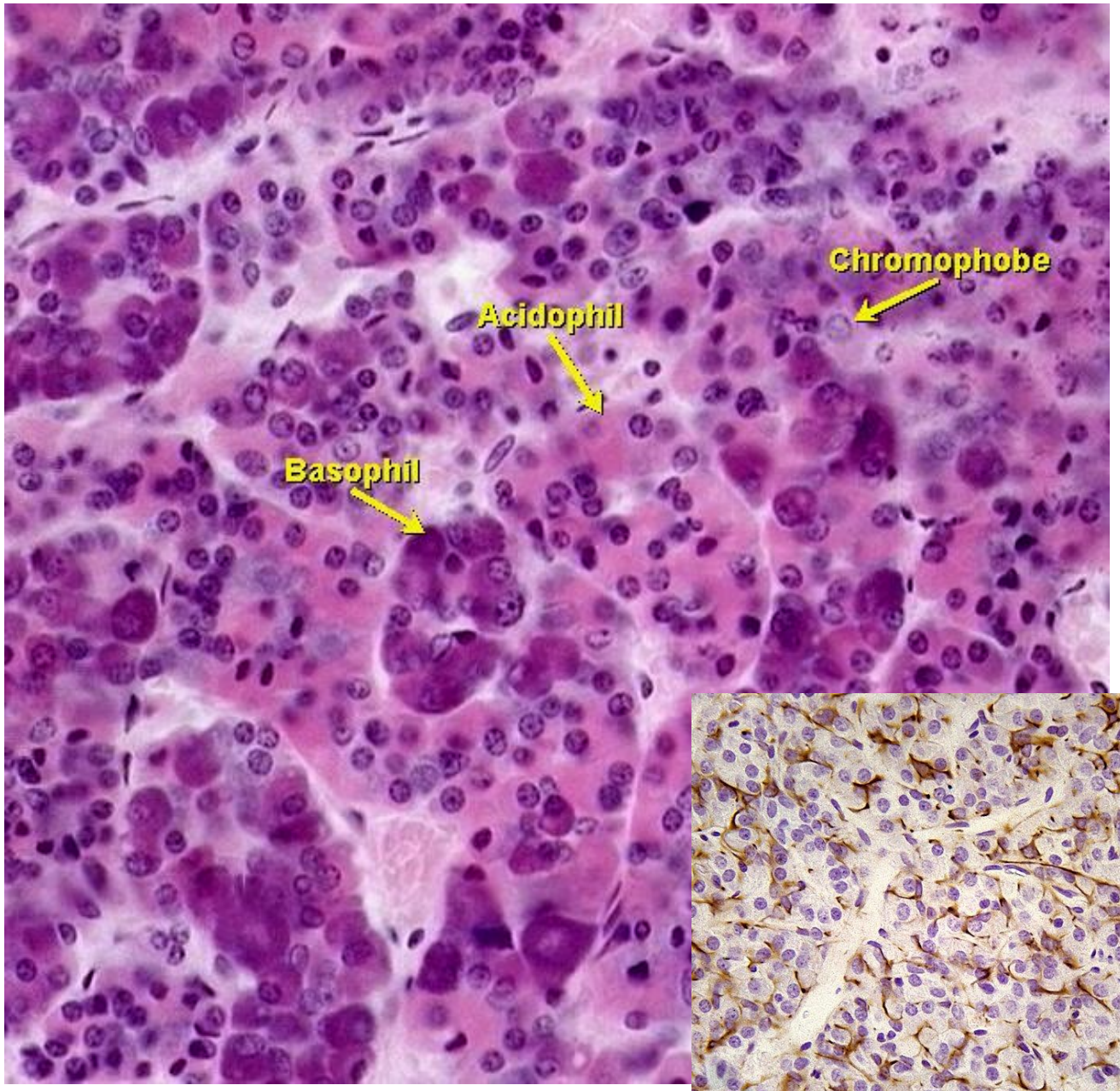
Pituitary

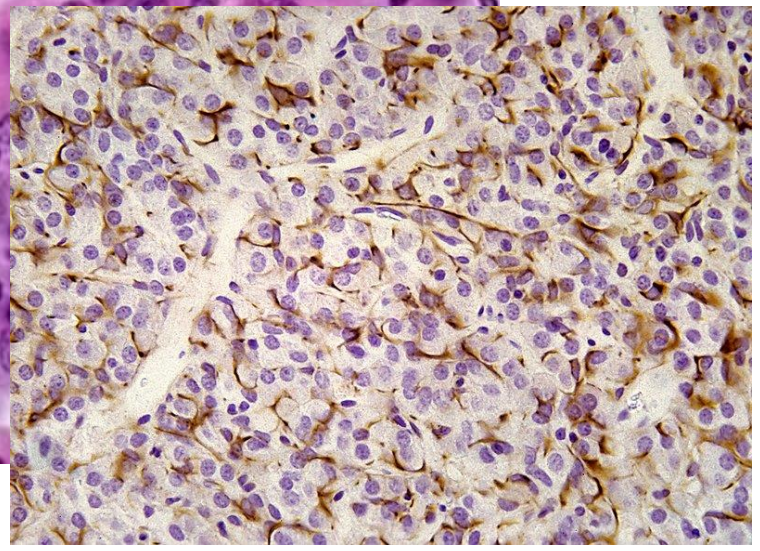
- Anterior lobe has three regions
  - Pars Distalis
  - Pars Intermedia
  - Pars Tuberalis
- Posterior/Pars Nervosa (neurohypophysis)
  - Attached to hypothalamus via infundibulum
  - Third ventricle may extend into neurohypophysis



Pars Distalis

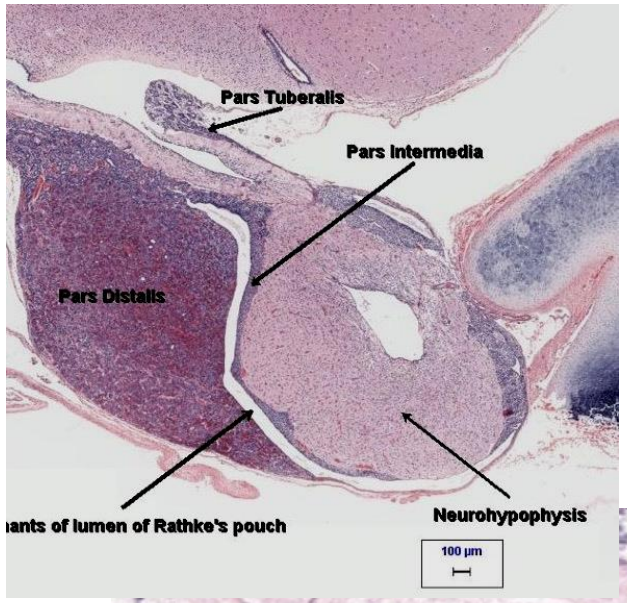
- Three cell types by staining
  - Basophils – basophilic cytoplasm
    - ACTH, LH/FSH, TSH
  - Acidophils – Eosinophilic cytoplasm
    - GH, PRL
  - Chromophobes – neutral cytoplasm
    - Post release?
- Folliculostellate cells
  - No hormones
  - Long processes
  - Gap junctions for communication

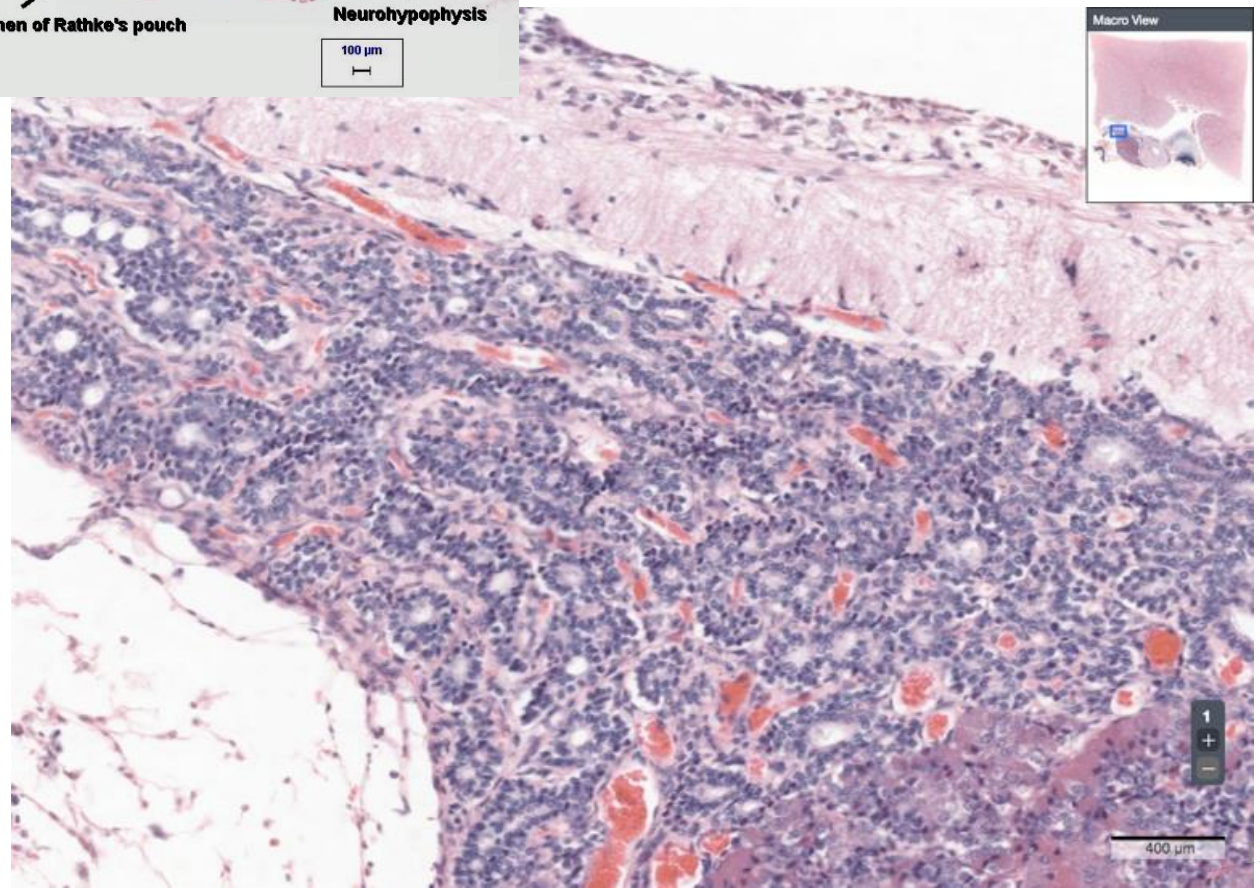




Pars Intermedia and Tuberalis

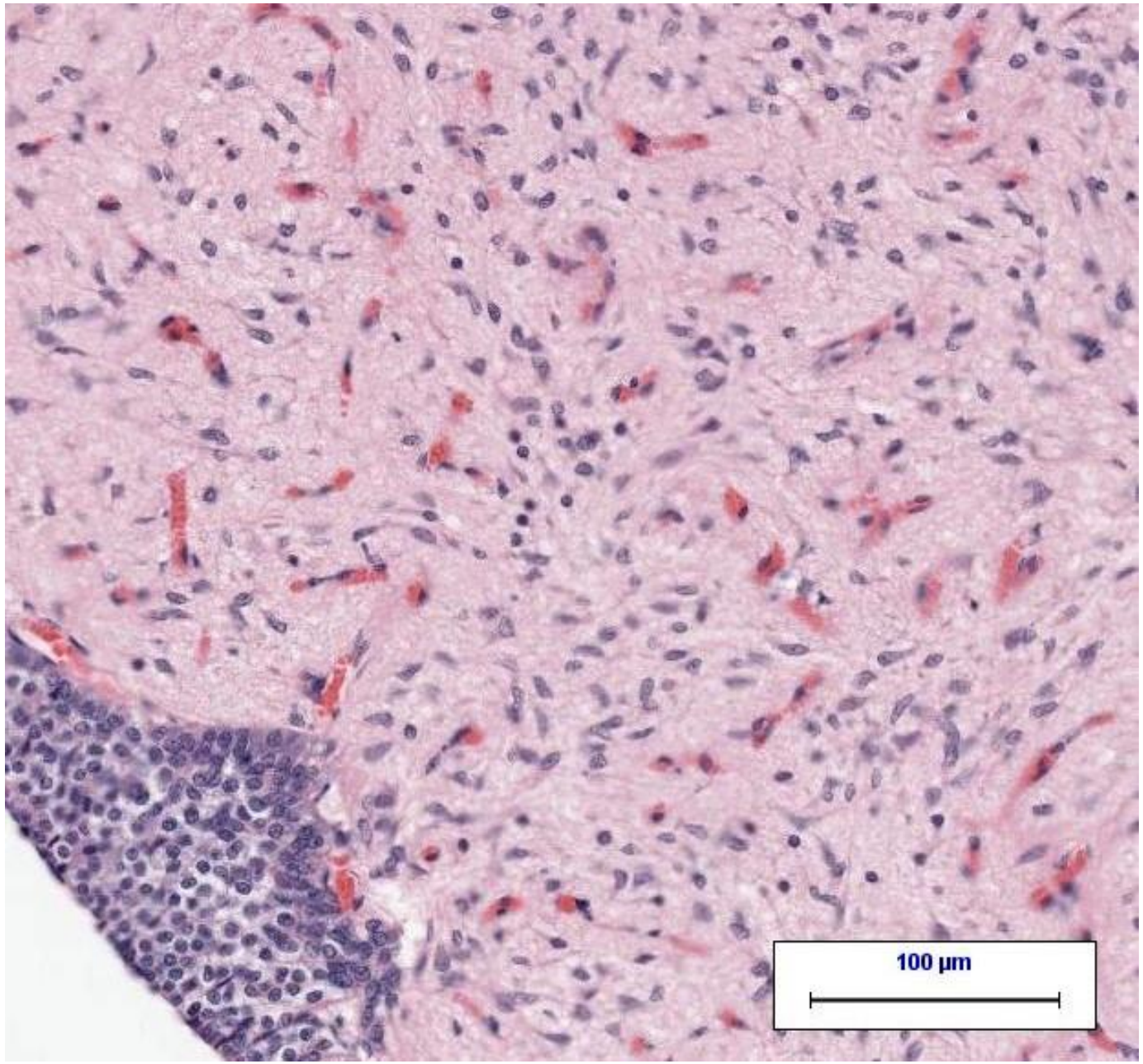
- Intermedia is adjacent to nervosa
  - Rathke's pouch remnants as colloidal cysts
  - Cells thought to be corticotropes
- Tuberalis wraps around the infundibulum
- Hypophyseal portal veins
- Epithelioid clusters
  - IHC staining shows cells positive for ACTH, FSH, & LH





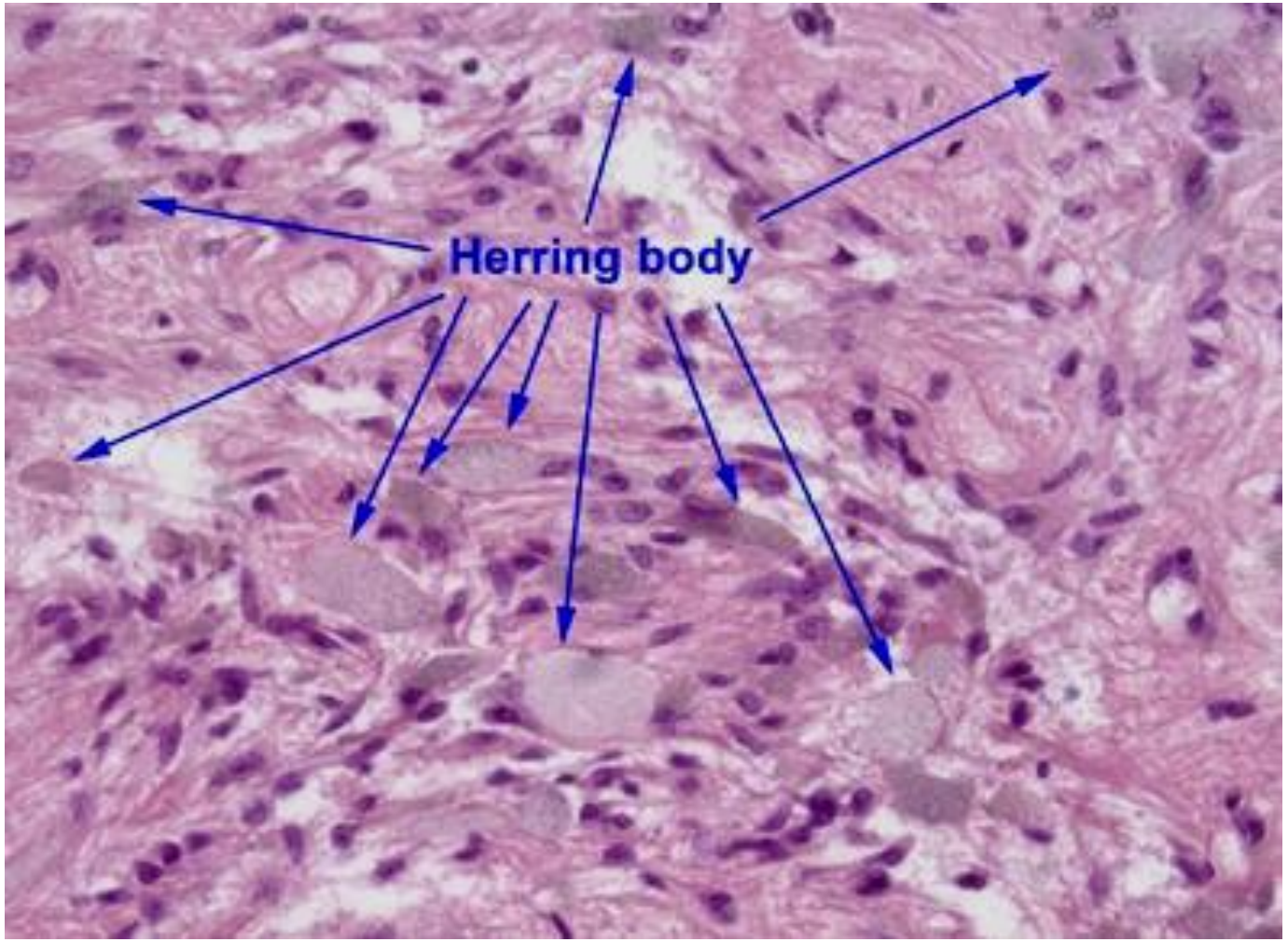
Pars Nervosa

- Major cell type is pituicytes
  - Glial like cells
- Unmyelinated axons of cells housed in hypothalamus
  - Supraoptic nuclei
  - Paraventricular nuclei
- Lots of Capillaries



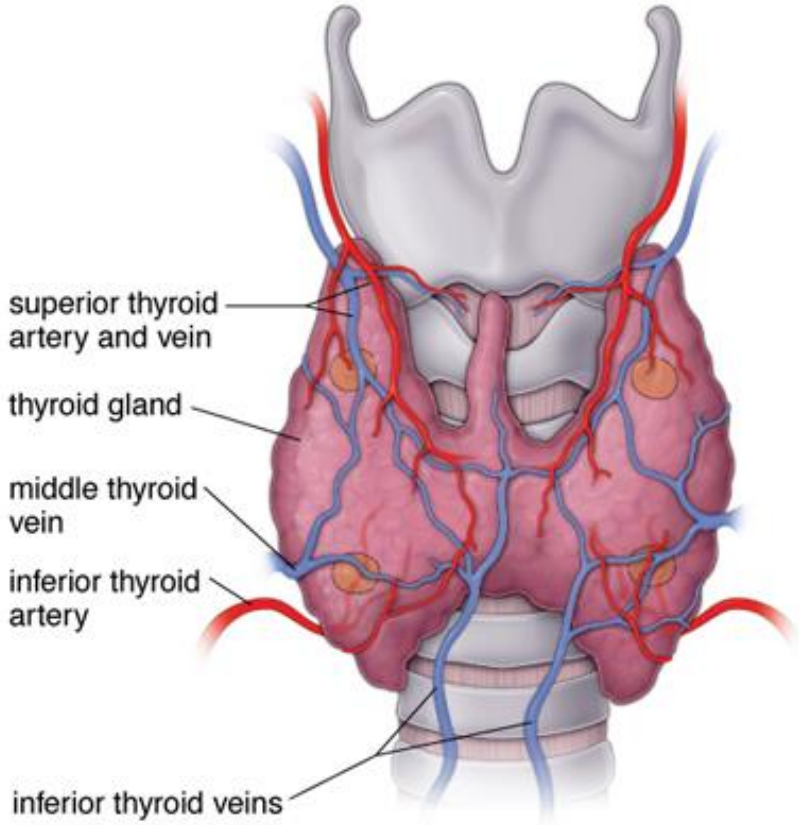
Herring Bodies

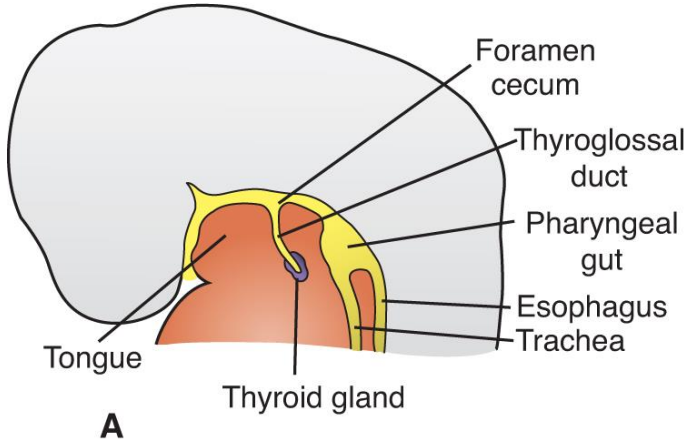
- Engorged axon terminals within the Pars nervosa
  - Oxytocin
  - ADH
- Will release if neurons in the hypothalamus are triggered

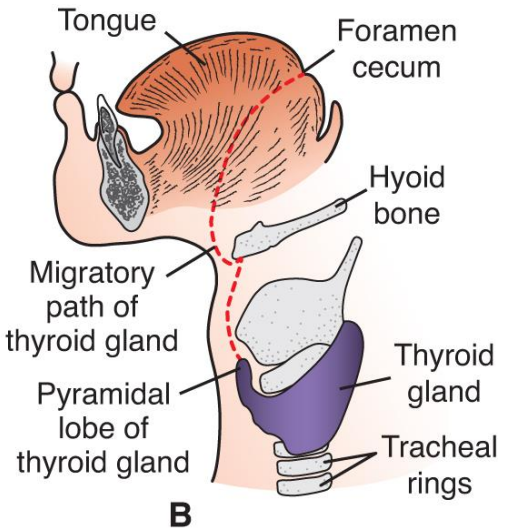


Thyroid Gland

- Encapsulated bi-lobed gland
  - Anterior neck
  - Isthmus
  - Pyramidal lobe
- From the pharyngeal endoderm via the thyroglossal duct
  - Foramen cecum
  - Thyroglossal duct will eventually obliterate
- Thyroid and calcitonin

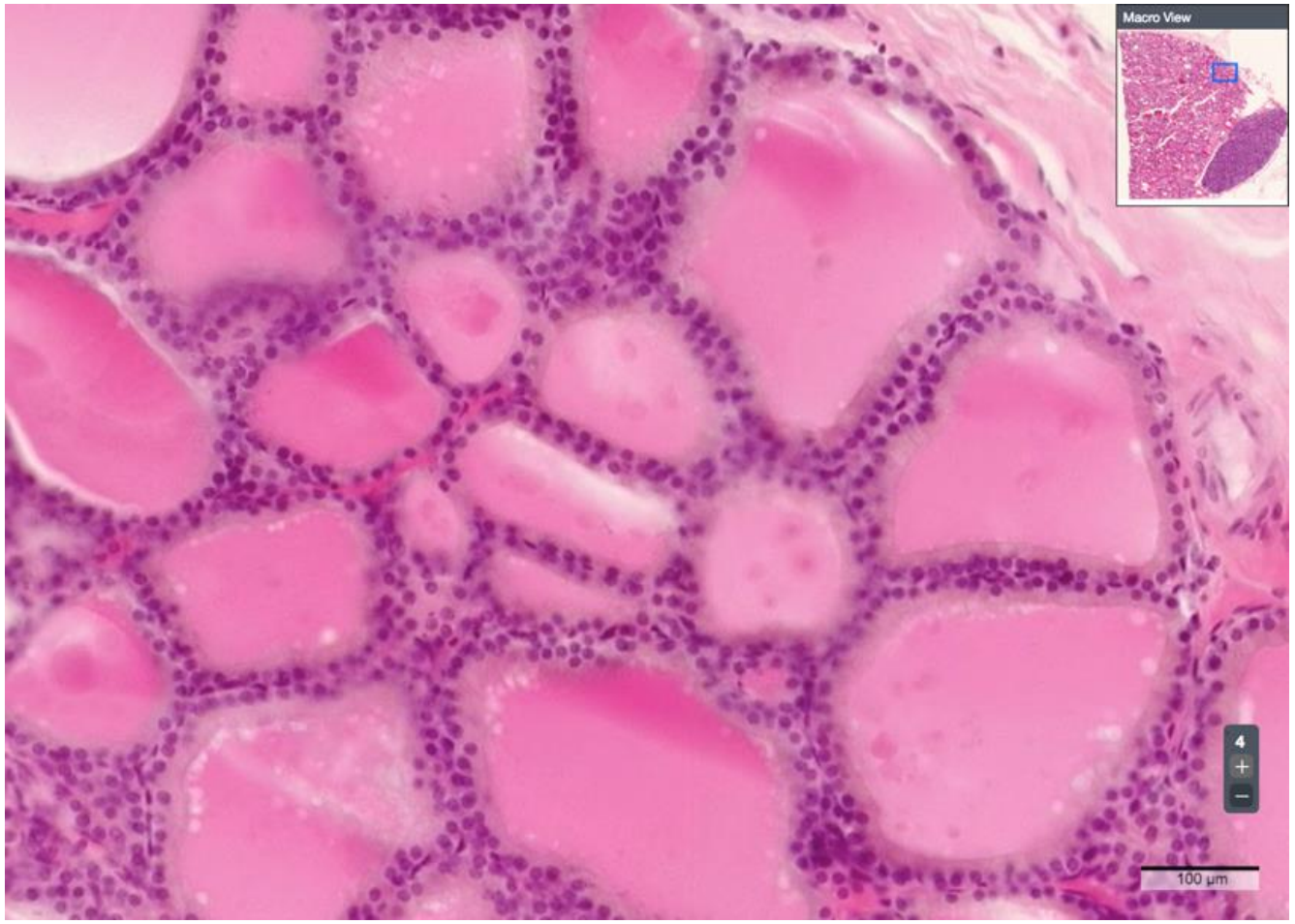






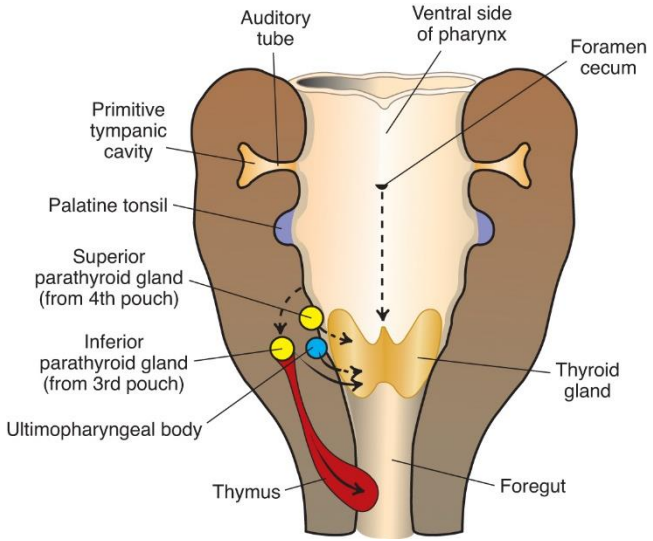
Follicles

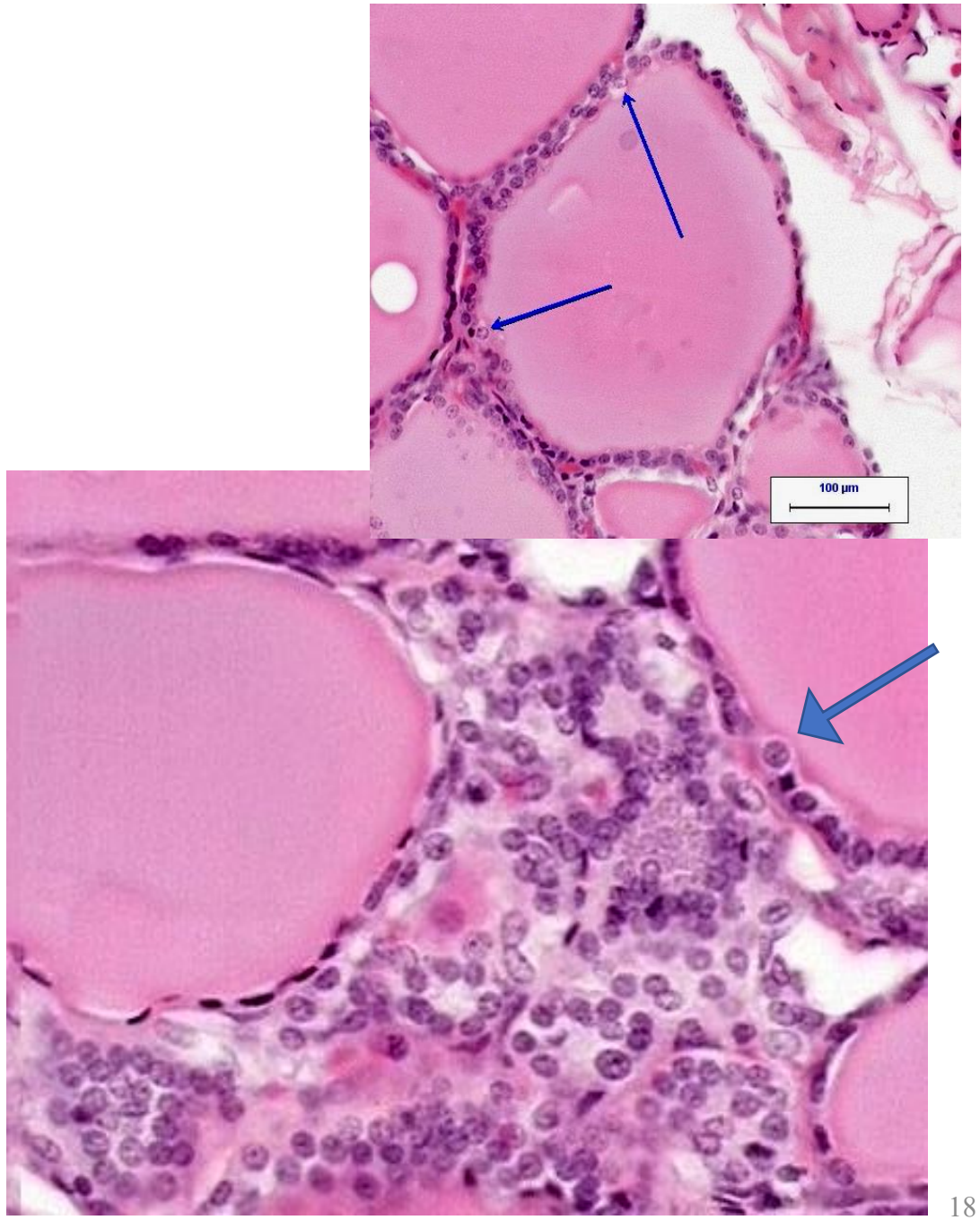
- Principal cells
  - Follicular Epithelium
  - Makes and resorbs colloid
- Colloid
  - Inactive hormone thyroid hormone
  - Thyroglobulin
- Shape of follicular epithelial cells suggests activity of the follicle
  - Range from very low cuboidal to columnar
  - Apical endocytic vesicles



Parafollicular/C-cells

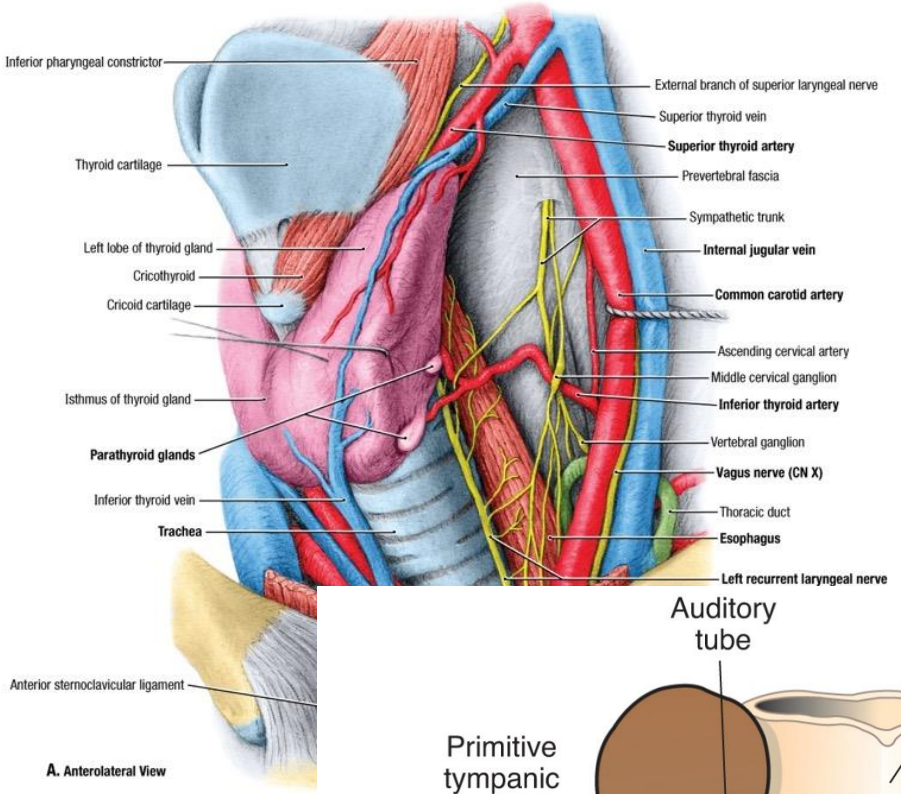
- Secrete calcitonin
- Reside in the follicular epithelial layer
  - Clearer cytoplasm
- Migrate in from the 4th pharyngeal pouch
  - From a region called the Ultimobranchial body

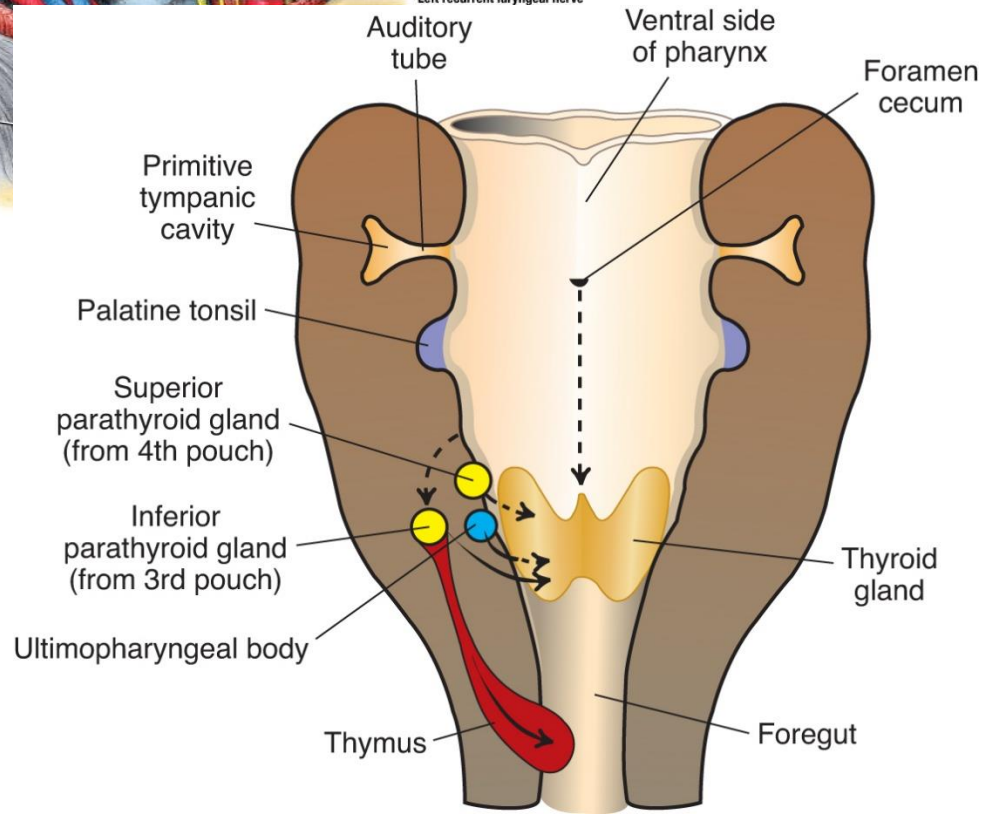




Parathyroid Glands

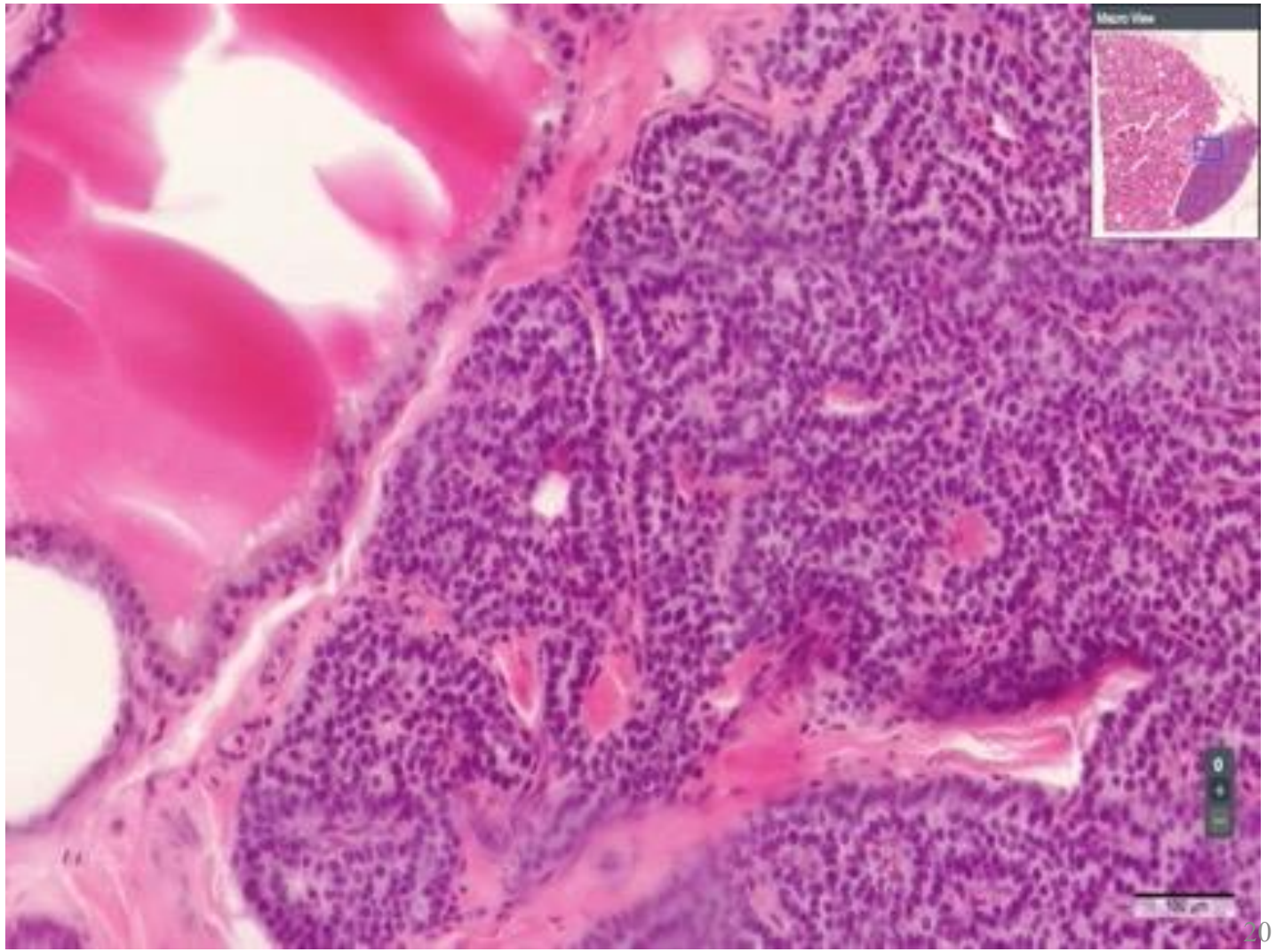
- Small encapsulated
  - Dorsal surface of thyroid
  - 2/lobe
- From 3rd and 4th pharyngeal pouches
  - Variable final positions
- Parathyroid Hormone
- Two cell types by histology
  - Chief/ principal cells
  - Oxyphils increase with age





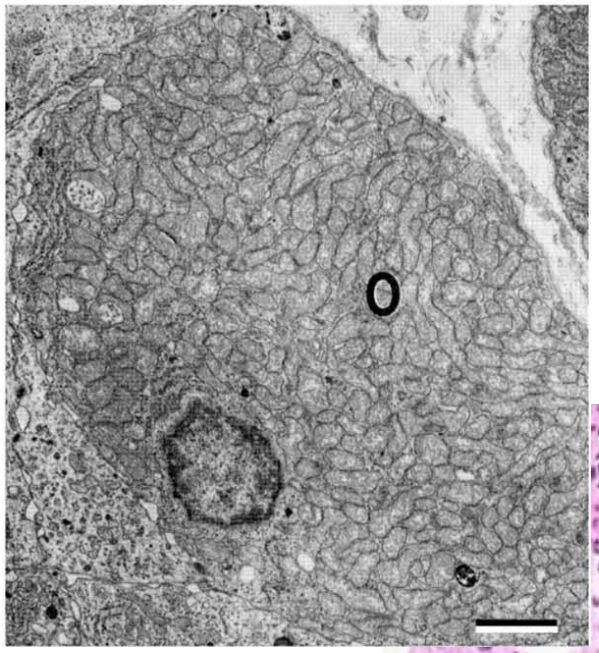
Principle/Chief Cells

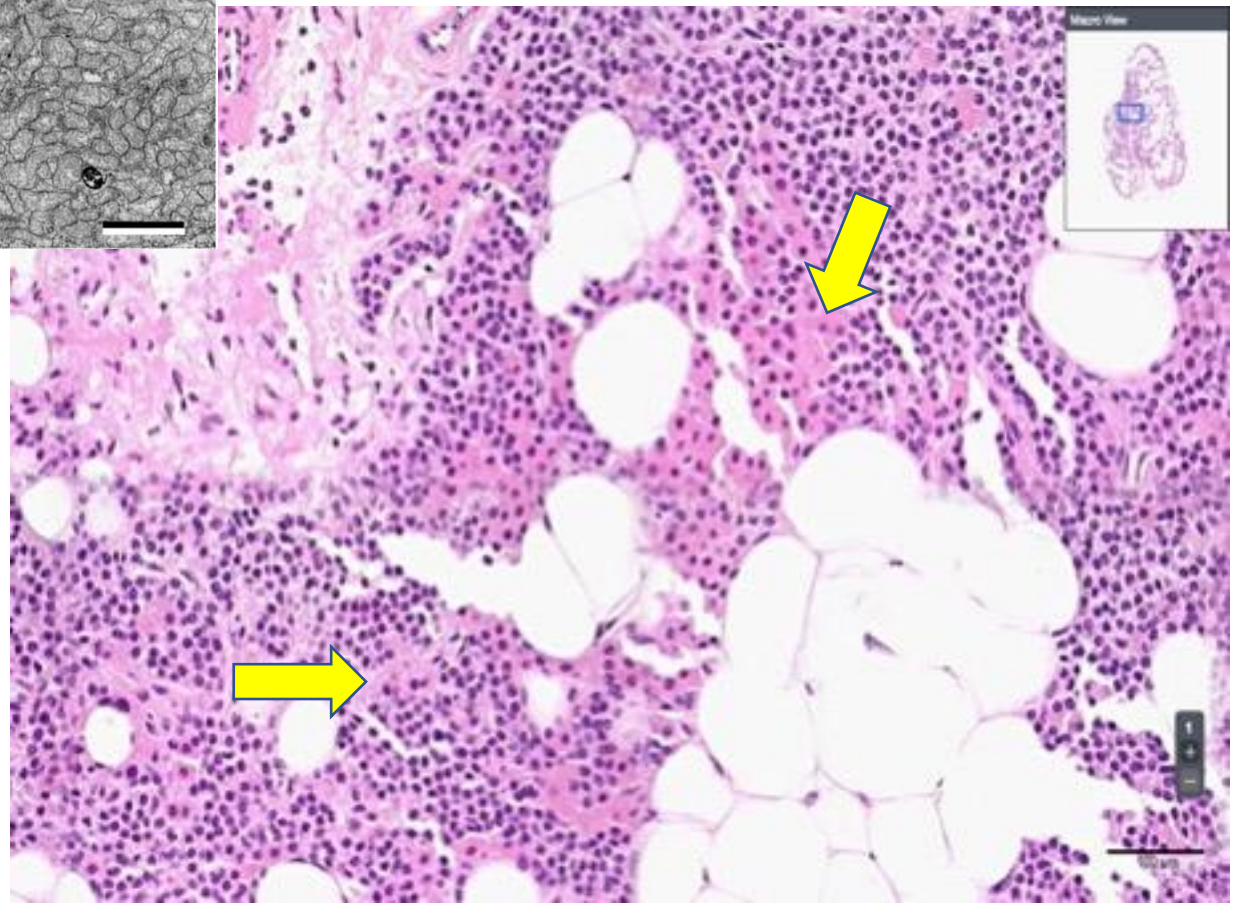
- Glandular epithelium
  - Cords and nests
  - Small polygonal
- Make Parathyroid hormone (PTH)
- Abundant fenestrated capillaries



Oxyphil Cells

- Larger cytoplasmic volume
  - Eosinophilic cytoplasm
  - Mitochondria
  - Can make hormone
  - Calcium receptors
  - Doing????
- Increase numbers with age
  - Oxyphil cell #'s increase in uremic secondary hyperparathyroidism
- Adipose increases with age

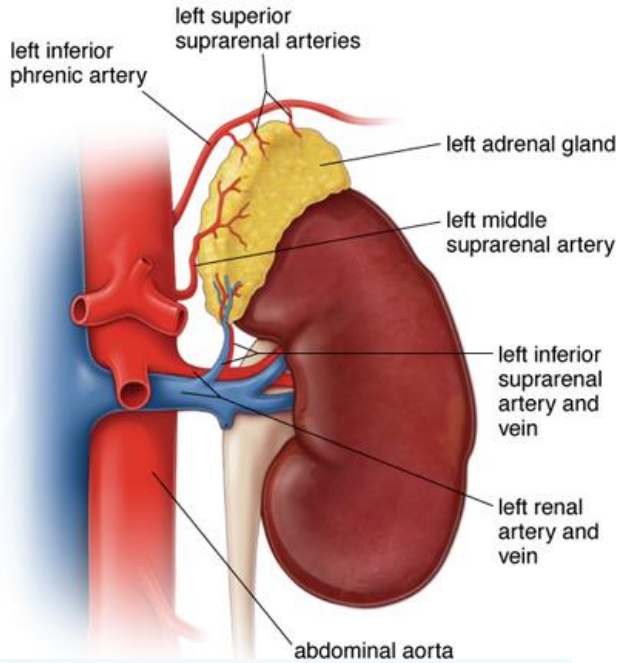


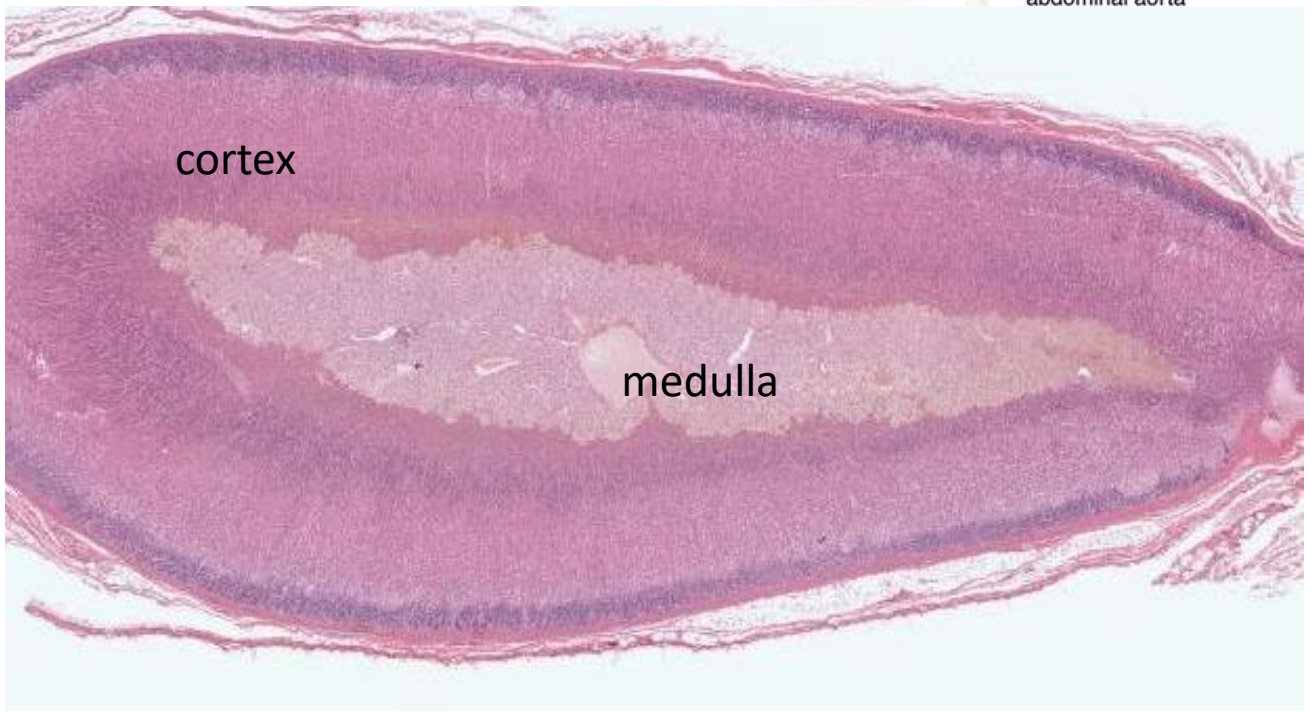


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Suprarenal/Adrenal Gland

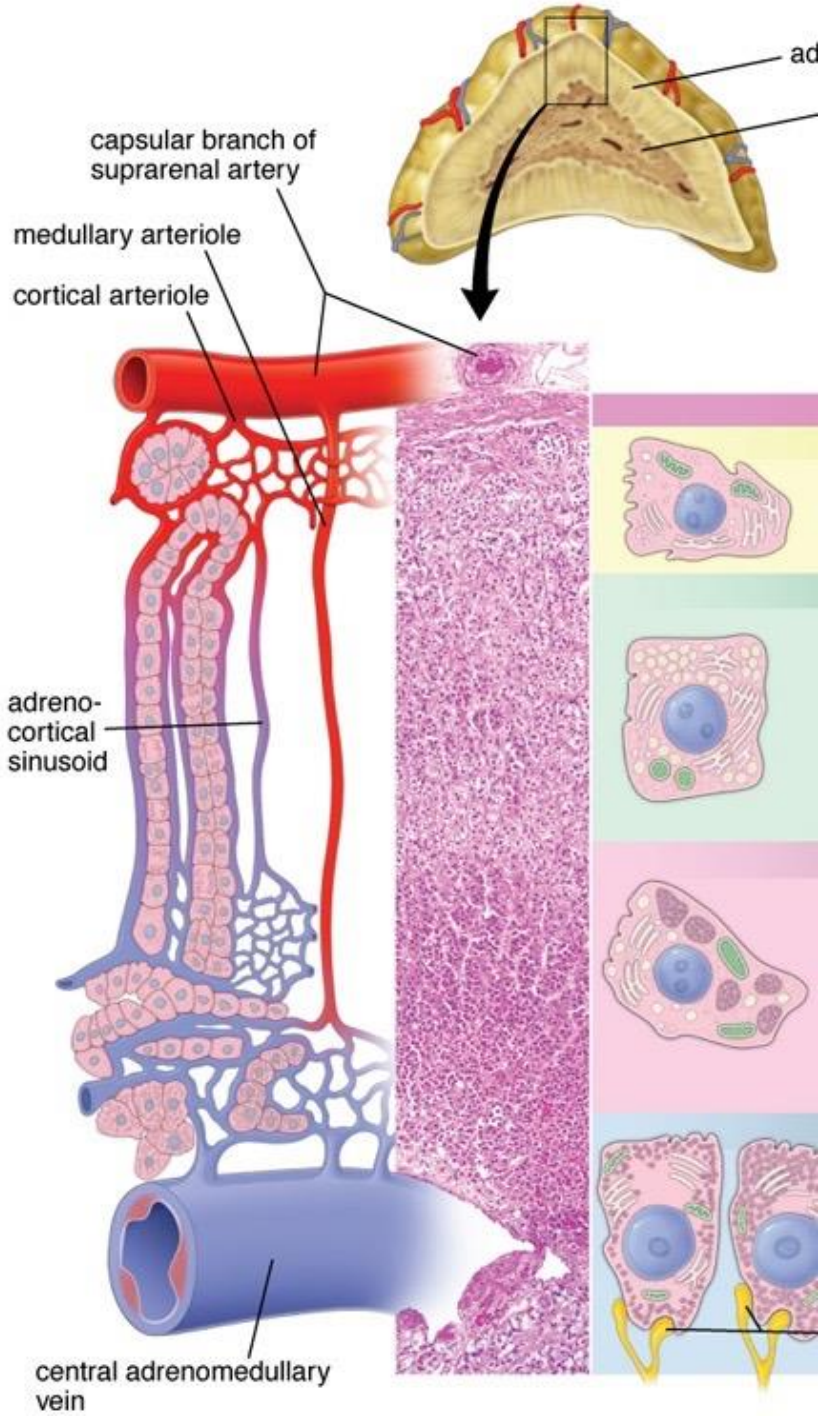
- Connective tissue capsule
- Cortex
  - Three layers of cells derived from intermediate mesoderm
- Medulla
  - Chromaffin cells from neural crest
  - Giant sympathetic ganglion
    - Acetylcholine from preganglionic fibers trigger the release of Epinephrine and Norepinephrine directly to blood





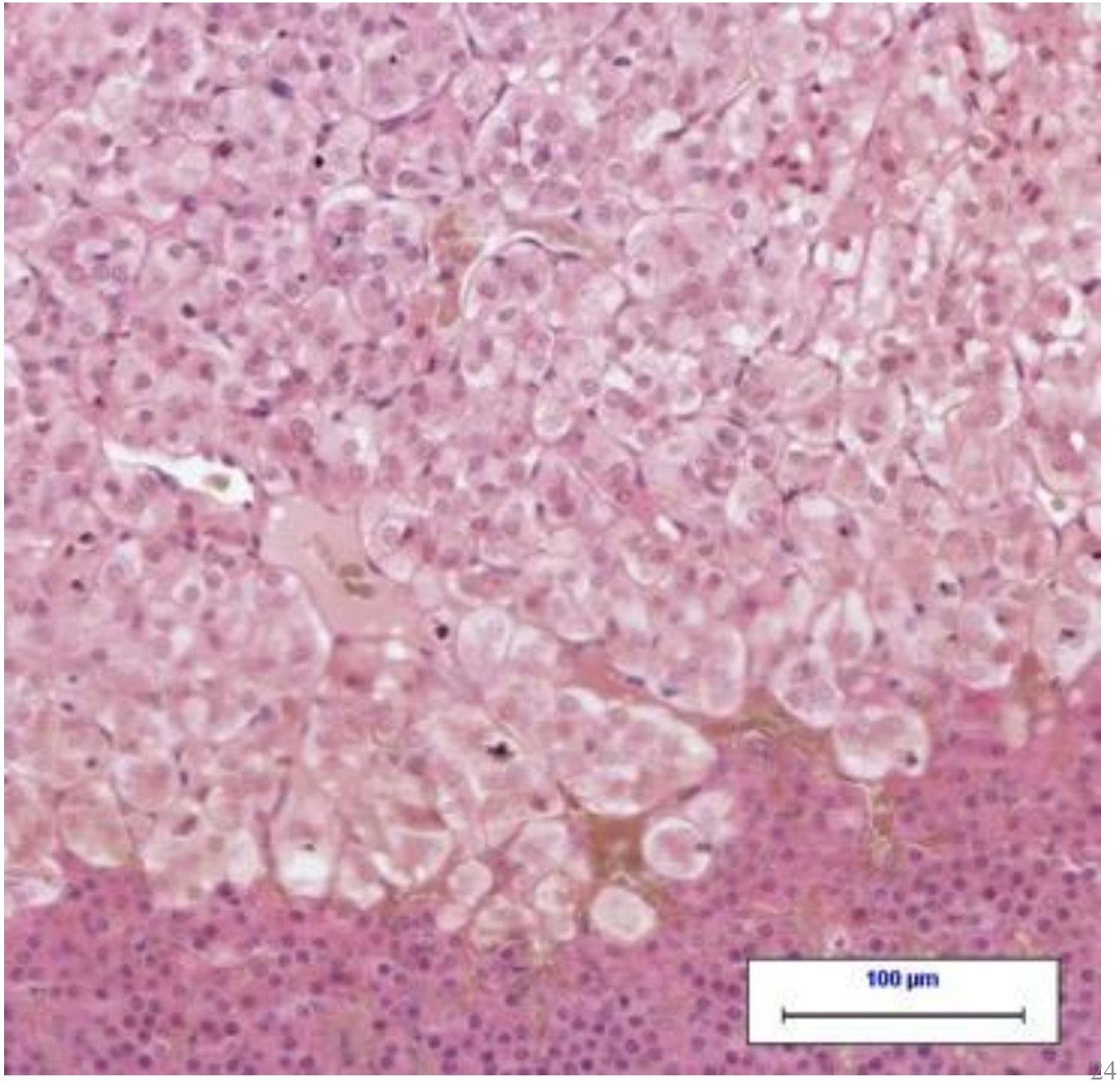
Blood Flow Through Adrenal Gland

- Capsular arteries send off dual blood supplies
- Cortical arterioles
  - Anastomosing network of fenestrated sinusoidal vessels in cortex
  - Drain into and through medullary cells
- Medullary arterioles
  - Bypass cortex to supply medulla
- One central vein



Adrenal Medulla

- Chromaffin cells
  - Epithelioid secretory cells
- Large with numerous secretory vesicles
  - Pale staining
- Catecholamines



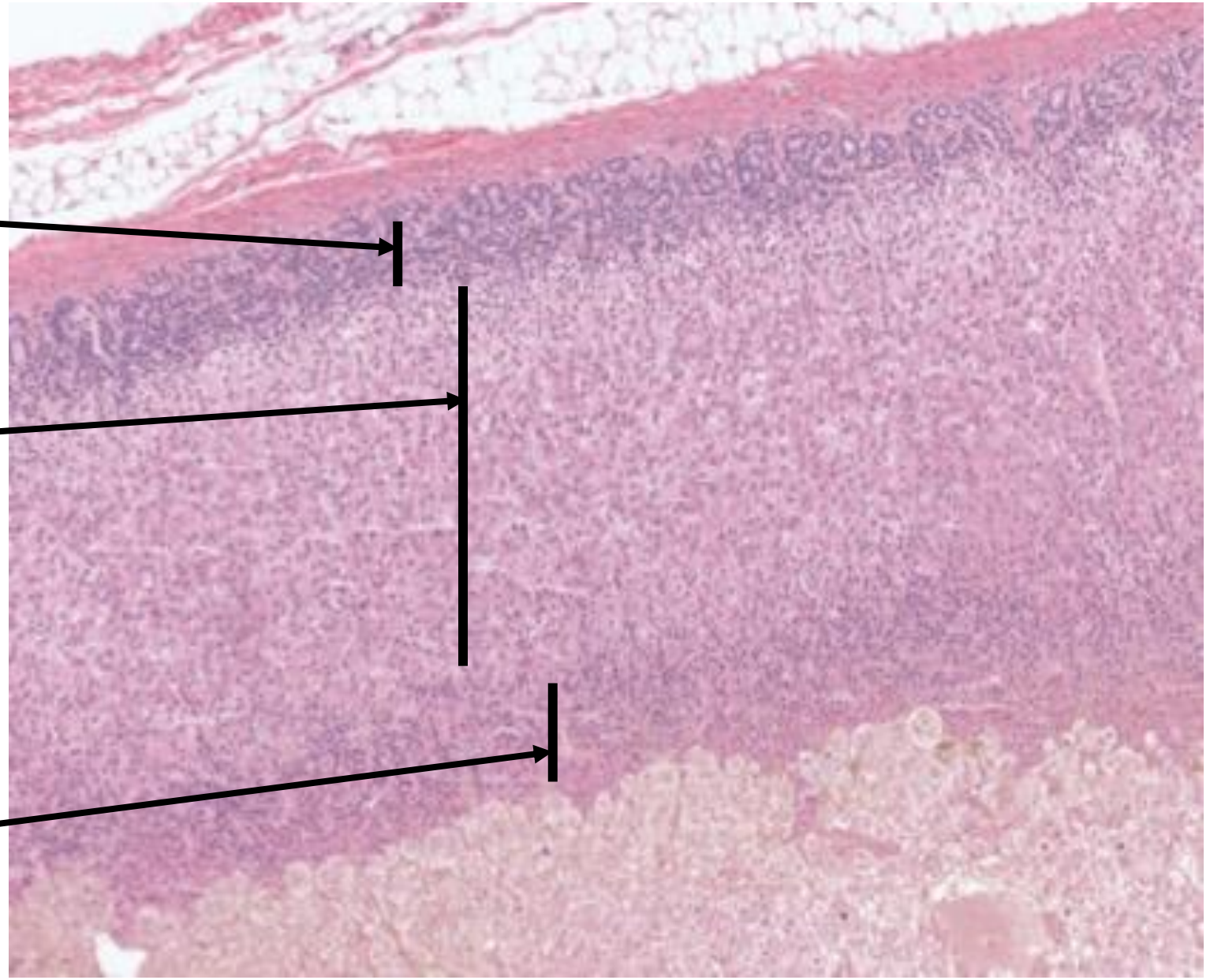
Adrenal Cortex

Three separate zona

Glomerulosa

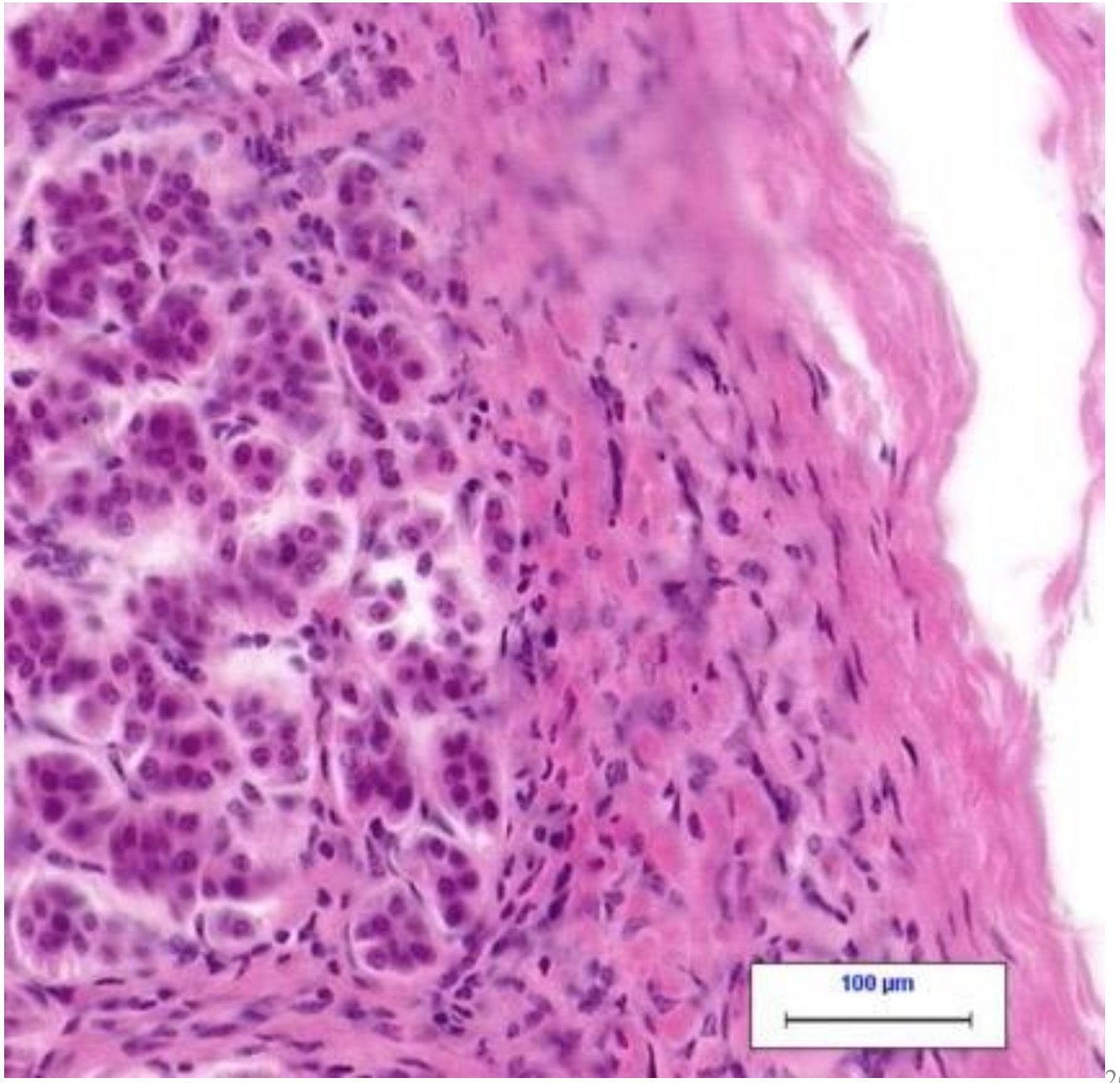
Fasciculata

Reticularis



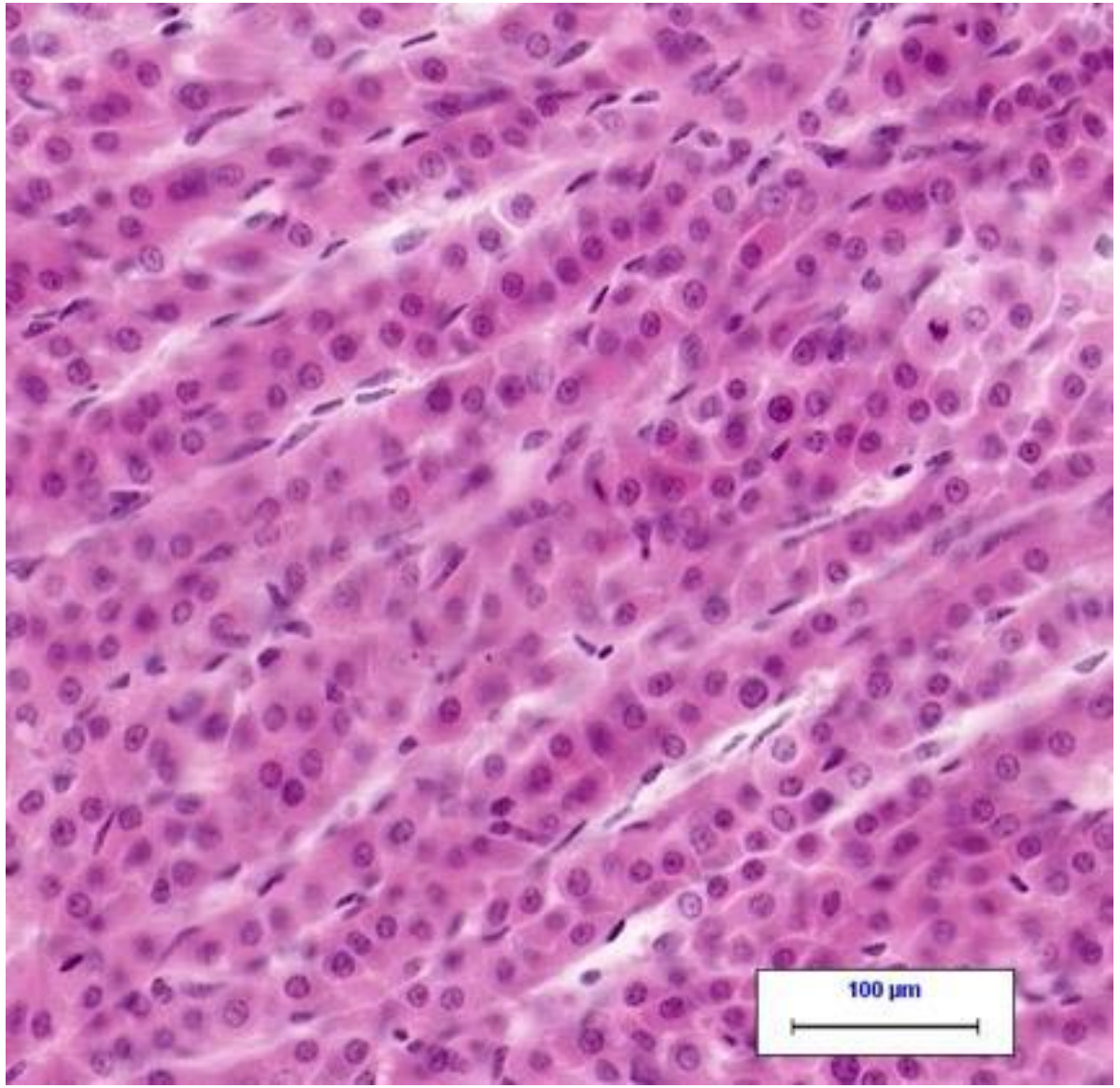
Zona Glomerulosa

- Narrow near capsule
- Clusters of cells
- Densely packed nuclei
- Secrete mineralocorticoids
  - Aldosterone



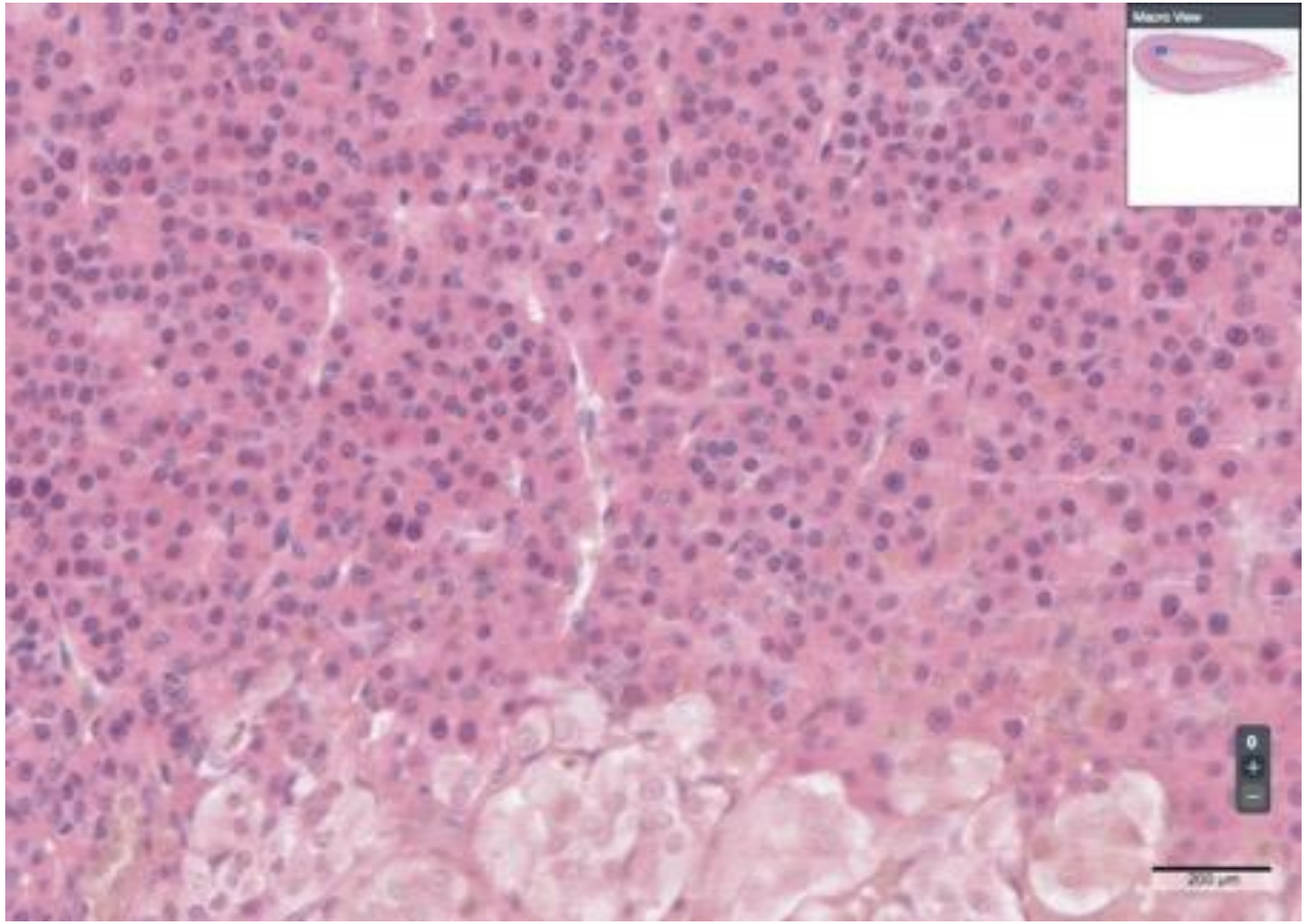
Zona Fasciculata

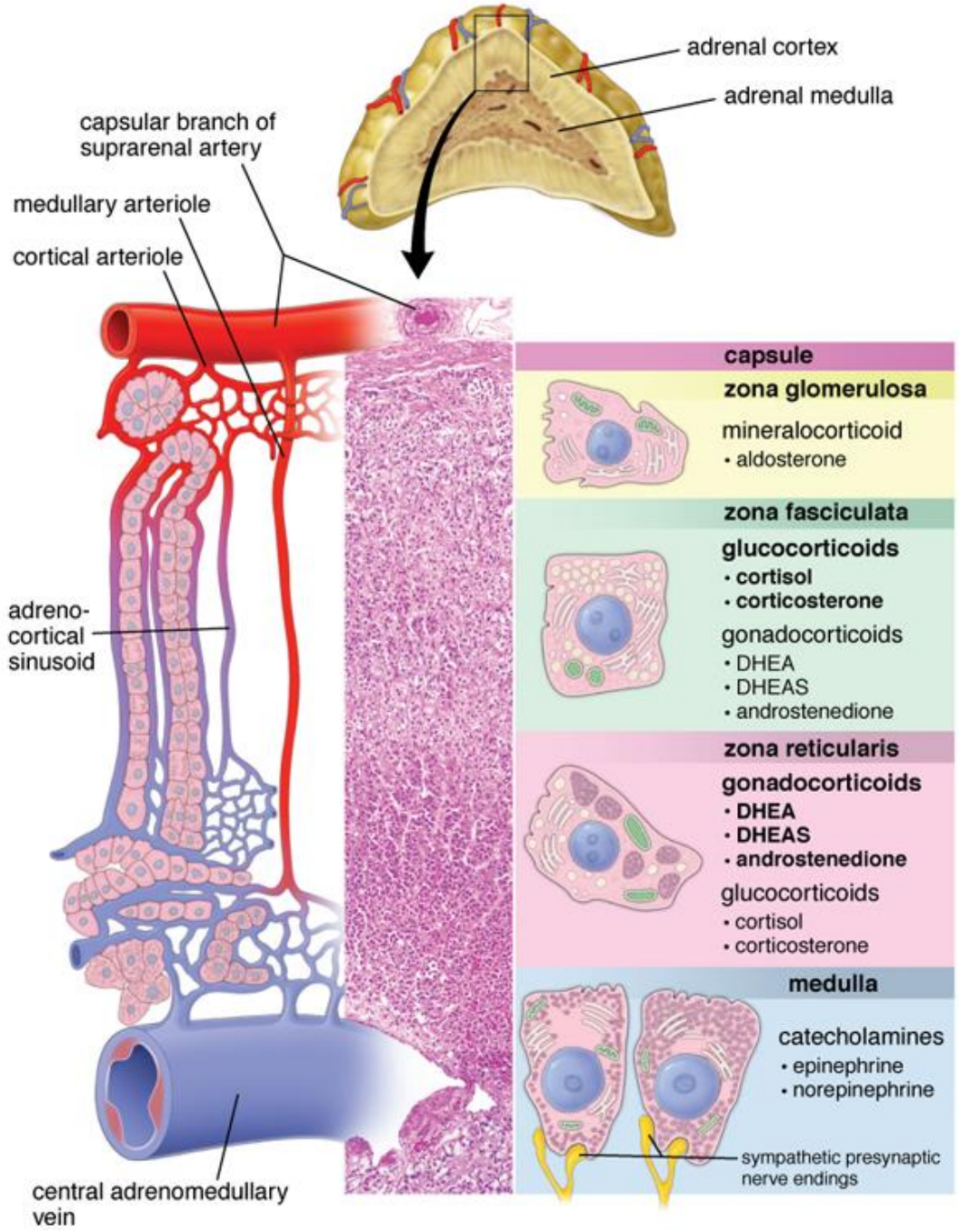
- Middle zone
- Long cords of cells
  - Larger
  - Paler nuclei
  - More acidophilic cytoplasm
- Sinusoidal capillaries
- Cortisol and corticosterone
  - Some gonadocorticoids



Zona Reticularis

- Deepest zone
- Short anastomosing cords
  - Small dark nuclei
- Fenestrated capillaries
- Gonadocorticoids
  - Some glucocorticoids




adrenal cortex
adrenal medulla
capsular branch of suprarenal artery
medullary arteriole
cortical arteriole
adreno-cortical sinusoid
central adrenomedullary vein
capsule
zona glomerulosa
mineralocorticoid
• aldosterone
zona fasciculata
glucocorticoids
• cortisol
• corticosterone
gonadocorticoids
• DHEA
• DHEAS
• androstenedione
zona reticularis
gonadocorticoids
• DHEA
• DHEAS
• androstenedione
glucocorticoids
• cortisol
• corticosterone
medulla
catecholamines
• epinephrine
• norepinephrine
sympathetic presynaptic nerve endings